

Python Symbols & Operators

Student Cheat Sheet — Names, meanings, and common uses

1. Basic Symbols / Punctuation

Symbol	Name	Common use / example
()	Parentheses / Round brackets	Function calls, grouping expressions, tuples
[]	Square brackets	Lists, indexing, slicing
{ }	Curly brackets / Braces	Dictionaries and sets
:	Colon	if/for/while blocks, functions, slicing, dictionaries
,	Comma	Separating values/elements
.	Dot / Period	Accessing methods and attributes
;	Semicolon	Separating statements on one line
' '	Single quotation marks	Defining strings
" "	Double quotation marks	Defining strings
''' ''' / """ """	Triple quotes	Multiline strings and docstrings
#	Hash / Number sign	Single-line comment
\	Backslash	Escape characters
_	Underscore	Variable names and special Python conventions

2. Arithmetic Operators

Symbol	Name	Common use / example
+	Plus / Addition	$10 + 5 \rightarrow 15$
-	Minus / Subtraction	$10 - 5 \rightarrow 5$
*	Asterisk / Multiplication	$10 * 5 \rightarrow 50$
/	Forward slash / Division	$10 / 5 \rightarrow 2.0$
//	Floor division	$10 // 3 \rightarrow 3$
%	Modulus / Modulo	$10 \% 3 \rightarrow 1$
**	Exponent / Power	$2 ** 3 \rightarrow 8$

3. Comparison Operators

Symbol	Name	Common use / example
==	Equal to	$a == b$
!=	Not equal to	$a != b$
>	Greater than	$a > b$
<	Less than	$a < b$
>=	Greater than or equal to	$a >= b$
<=	Less than or equal to	$a <= b$

4. Assignment Operators

Symbol	Name	Common use / example
=	Assignment	$x = 10$

Symbol	Name	Common use / example
<code>+=</code>	Addition assignment	<code>x += 5</code>
<code>-=</code>	Subtraction assignment	<code>x -= 5</code>
<code>*=</code>	Multiplication assignment	<code>x *= 5</code>
<code>/=</code>	Division assignment	<code>x /= 5</code>
<code>//=</code>	Floor division assignment	<code>x //= 5</code>
<code>%=</code>	Modulus assignment	<code>x %= 5</code>
<code>**=</code>	Exponent assignment	<code>x **= 5</code>

5. Logical Operators

Symbol	Name	Common use / example
<code>and</code>	Logical AND	<code>a > 10 and b < 20</code>
<code>or</code>	Logical OR	<code>a > 10 or b < 20</code>
<code>not</code>	Logical NOT	<code>not(a > 10)</code>

6. Identity Operators

Symbol	Name	Common use / example
<code>is</code>	Identity	Checks whether two variables refer to the same object
<code>is not</code>	Identity NOT	Checks whether two variables do not refer to the same object

7. Membership Operators

Symbol	Name	Common use / example
<code>in</code>	Membership	"Python" in languages
<code>not in</code>	Not membership	"Java" not in languages

8. Bitwise Operators

Symbol	Name	Common use / example
<code>&</code>	Bitwise AND	Works at the bit level
<code> </code>	Bitwise OR	Works at the bit level
<code>^</code>	Bitwise XOR	Works at the bit level
<code>~</code>	Bitwise NOT	Inverts bits
<code><<</code>	Left shift	Shifts bits to the left
<code>>></code>	Right shift	Shifts bits to the right

9. Special / Common Python Symbols

Symbol	Name	Common use / example
<code>@</code>	At symbol	Decorators
<code>-></code>	Arrow	Function return type annotation
<code>*</code>	Asterisk	Unpacking / variable arguments
<code>**</code>	Double asterisk	Dictionary unpacking / keyword arguments
<code>:=</code>	Walrus operator	Assignment expression

Symbol	Name	Common use / example
...	Ellipsis	Placeholder / advanced Python usage

■ Beginner Must-Know Symbols

Students starting Python should first be comfortable with these:

Symbol	Name	Common use / example
()	Parentheses	Function calls and grouping
[]	Square brackets	Lists, indexing, slicing
{ }	Curly brackets / Braces	Dictionaries and sets
:	Colon	Blocks, slicing, dictionaries
,	Comma	Separating values
.	Dot	Methods and attributes
=	Assignment operator	Stores a value in a variable
==	Equal to	Compares two values
!=	Not equal to	Compares two values
> / <	Greater / Less than	Comparisons
>= / <=	Greater/Less than or equal	Comparisons
+ - * /	Arithmetic operators	Basic calculations
//	Floor division	Division rounded down
%	Modulus	Remainder
**	Exponent / Power	Powers
#	Hash	Comment
' ' / " "	Quotes	Strings
_	Underscore	Naming conventions
@	At symbol	Decorators

■■ **Important:** = means assignment, while == means comparison/equality.

Teaching Tip: Learn the proper names of symbols. This makes it easier to read code, explain errors, and communicate with other developers.